

Twins weather and sensor network — a narrative

1. Background — why, where, and the first cameras

I've been building hardware and writing code since I was a teenager — RadioShack parts, TRS-80 in 1979, first paid program at twelve, then cryptography, telephony, sensor frameworks, and more Arduino projects than I can count. I grew up in Maine skiing in the 80s and snowboarding by the late 80s. I like connected gadgets and I like being outside; this project is where those two things collided.

When I moved here four or five years ago, I started exploring the Twins — North Twin, South Twin, Bowman Mountain, Dailey Prairie, and the ridges and drainages between them. I got a permit and kept driving out just to **see** what conditions were like. The Twins gave me a reason to keep showing up in a new place; first trips stuck with me for the terrain, the access logistics (weather windows, snow, gates), and the recurring wish for **eyes on conditions** without driving to the middle of nowhere every time.

Bowman Ridge is an informal name I use — Bowman Mountain has been labeled differently on maps over the years. The ridge runs from Bowman along the west side of the "bathtub" that is Dailey Prairie, toward Blue Mountain Road, and forms the Orsino Creek drainage toward Skookum. It's a strong sensor site: high enough to see Baker and the Twins, often with cell coverage facing town, though the highest points sit around 4,000 feet and stay buried in snow much of the year.

Valley forecasts and airport weather often don't match what's happening on the Twins.

Ridge microclimate — wind, temperature, snow cover, visibility — can be completely different from town or a consumer app. That gap is why local observations matter for backcountry skiing, route planning, and go/no-go calls in sketchy weather. For instance the weather on the SW side of Bowman Ridge is very different than the NE side.

After enough repeat drives, the question was obvious: *why am I doing this when I could put cameras and sensors up there?* So I bought a trail camera, put it on Bowman Mountain, and started watching. Then it got out of control — in a good way. Fire season pushed me toward **air quality on the ridges**, not just in town. Temperature and pressure came first, then snow depth, wind, soil probes, doors, and a LoRaWAN mesh to tie it together.

My friend Anders — who taught me to snowboard in the 80s — jokes that I combined woods time and harsh-environment electronics. Anders (electrical/computer engineer), Gabe, and Dave have helped with battery sizing, equations, and the electrical details I'm still learning. I carry a Starlink kit in the field without cell service — which is how part of this got written while driving out to un-brick a device at sunset, looking at North Twin and Baker with the dogs.

Trail cameras

There are **five trail cameras** on the project website right now — easier to spot than the small sensor nodes because they need a clear view. I installed the first ones **in the middle of winter**, including trips in freezing rain that were miserable. Placement targets terrain and conditions, not people. They have caught a few passersby anyway — I've written code to drop frames with people in them, and I'm moving cameras higher in trees and farther from trails where I can still get useful shots without being creepy.

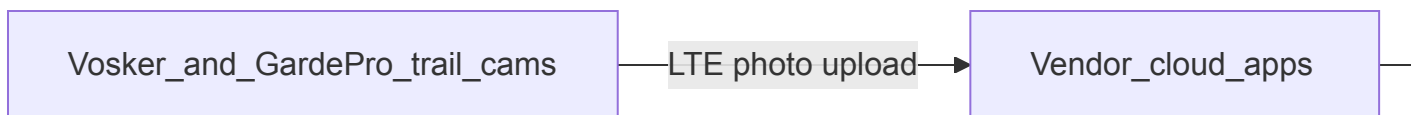
The camera story started with off-the-shelf **cellular trail cams**, not custom hardware. I run **Vosker** and **GardePro** units — battery-powered, LTE-connected, built for remote woods deployment. They wake on a schedule or motion trigger, upload photos to each vendor's cloud app, and I pull from there. With help from Claude, I wrote scraping code to get images onto the project website — a couple thousand frames so far, with a seasonal stop-motion in the works. Some cameras have been up since around October. **This is still what feeds the site today.**

Only later did I start on a **custom high-resolution trail camera**: 3D-printed enclosure, high-res UVC camera, Particle Tachyon single-board computer. Particle also gave me a satellite module I haven't deployed yet — uplink is expensive — but it would be useful on the far side of the Twins, maybe even near the glaciers, where there's no cell. That build is **in progress**; it's not what's running the four live cameras.

What's running now — Vosker and GardePro: Commercial cellular trail cameras. Each unit uploads over LTE to its vendor cloud (Vosker app / GardePro app). A scraper pulls new images from those services onto the project website. Simple, reliable, and what got the whole camera program started.

What's next — custom Particle Tachyon: Self-built camera with USB UVC sensor, LTE uplink, SD card buffer, and cloud storage at Backblaze. It can optionally publish temperature and pressure in the same format as the weather nodes. Woods Cam 001 is the first of these — still being field-tested alongside the commercial units. I now also have Woods Cam 002 which is rasp pi based.

Diagram — trail cameras today (Vosker / GardePro)



2. Particle.io fleet

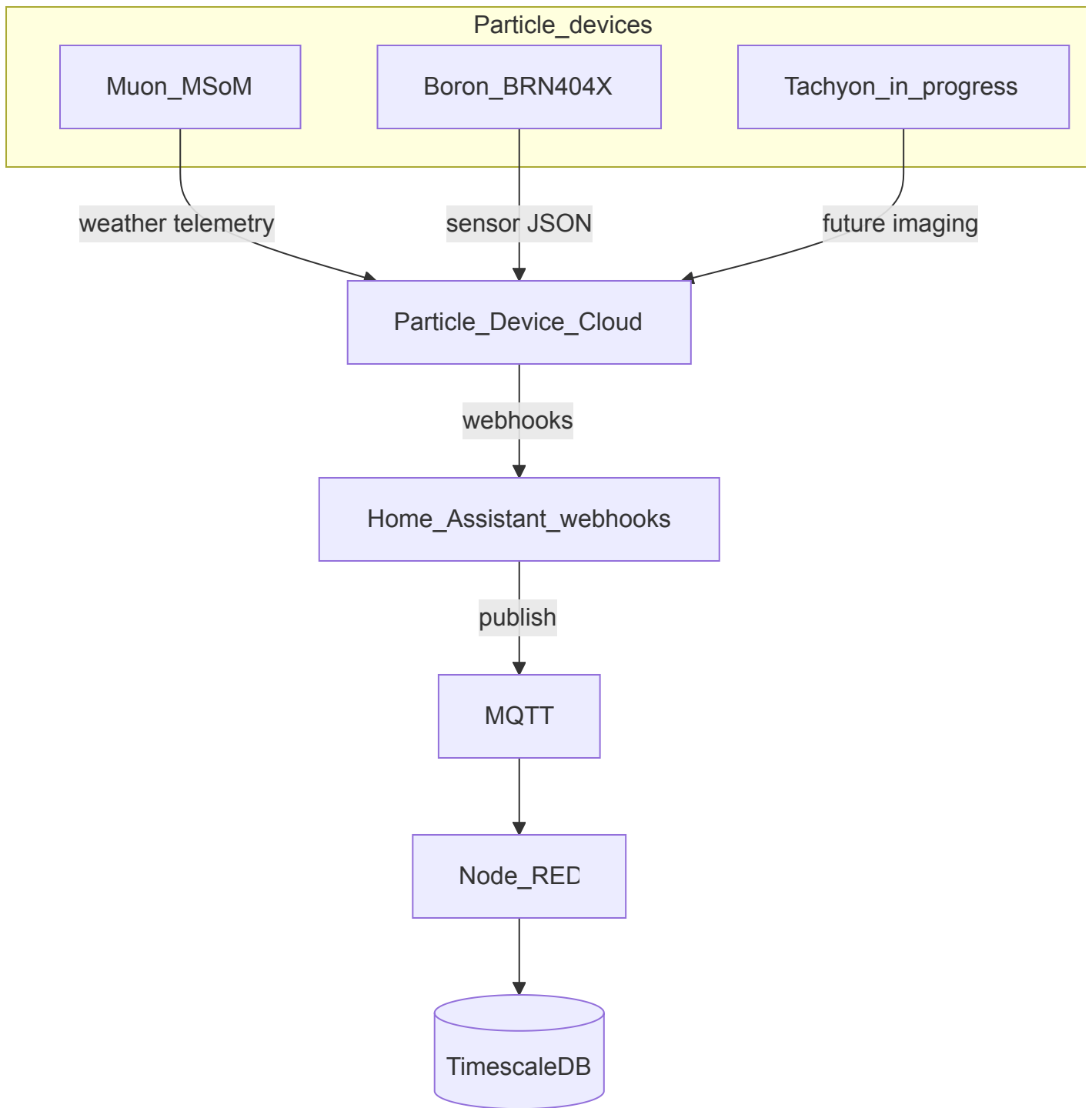
The first custom field sensor was a **Particle Boron** — ESP32-class board with built-in cellular — on Bowman. It worked, and it didn't: cellular up there is sketchy. That pushed two paths at once — **cellular** where coverage exists, **LoRaWAN** where it doesn't (or as backup).

I've since built **four custom air-quality sensors** on **Particle Muon** hardware (Wi-Fi, Ethernet, LoRaWAN, cellular — deploy with whatever fits the site). They use Sensirion particulate sensors over I2C, with a UART conversion step for an address conflict on the bus. They're more noticeable than Dragino nodes and expensive — roughly \$500–600 each in parts, not counting hours. I built them as a labor of love during fire season; most parts were on hand, and an LLC absorbs some project costs.

Most Muons and Borons publish to **Particle Device Cloud**. **Home Assistant** receives webhooks and forwards normalized traffic into **Node-RED**, which persists rows into **TimescaleDB**.

The Particle side of the fleet includes **Muon weather and air-quality units**, **Boron weather sensors**, and the **Tachyon trailcam** (in progress — see §1).

Diagram — Particle → Home Assistant → database



3. LoRaWAN and The Things Network

LoRaWAN is a long-range, low-power radio protocol built for distributed environmental monitoring. In practice, what surprised me is how **promiscuous** the network is: put up a gateway and you extend the mesh. My sensors hear other people's traffic — I'm picking up meshes along Mosquito Lake Road — and vice versa.

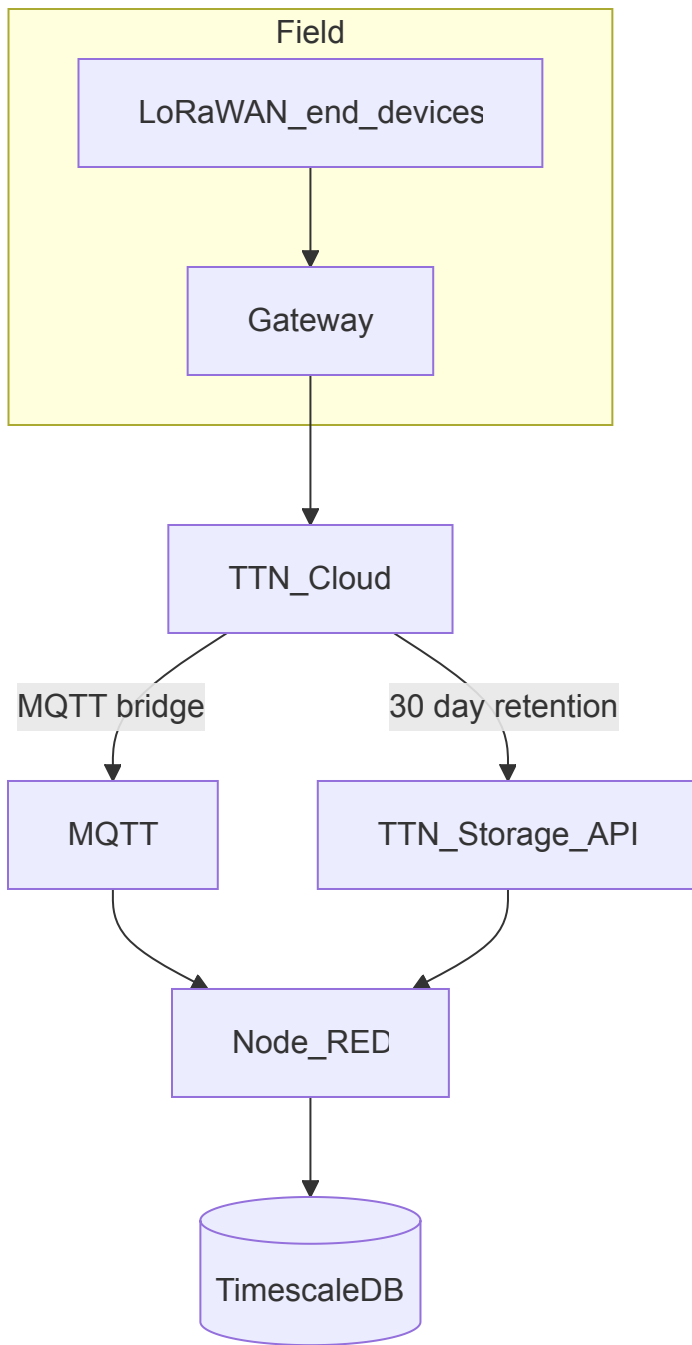
The main woods gateway sits on **North Twin** — solar powered, **cellular backhaul**. Winter taught me that **batteries really don't like cold**: even with what I thought was enough solar and battery headroom, cold snaps shut the gateway down. This summer I'm hauling up **heated battery** setups and other lessons from a season of alpine winter ops.

End devices are small — roughly a third of a loaf of bread, sometimes smaller. Most gear has QR-coded tags; scan one and you'll land on the project site. I guess I accidentally built **geocaching**. I hope that's fun rather than annoying.

I'm also considering gateways along **Bowman Ridge** — more accessible than North Twin, though North Twin has the elevation advantage. Still thinking that through.

Payloads go through the gateway to **The Things Network** (TTN), which bridges into MQTT at my house, where they're decoded and normalized to the same shape as Particle data before landing in the database.

Diagram — LoRaWAN → gateway → TTN → database



4. What's in the field

Most nodes fall into two families:

1. **Dragino** — rugged, solar-powered, off-the-shelf environmental sensors (temp, humidity, snow depth, soil temp, doors, buttons, etc.). Built for ranches and distributed monitoring; a good fit for the woods.
2. **Custom Particle Muons** — air quality and anything needing more compute or multiple radios.

Beyond temp and pressure, the fleet now includes **three ultrasonic wind sensors** (one deployed on Bowman Ridge so far). They're not heated — winter behavior is TBD. The hope is that keeping them around 3,500–4,000 feet, where sun cycles warm them, will prevent rime and jamming. The goal is **differential wind**: one along the skin route on the **northwest face of North Twin**, maybe one on **South Twin**, maybe one on **Bowman**, to compare what the ridges are actually doing.

A fair amount of gear came from wholesale suppliers — samples, donations, and "please sell me one unit" emails. It's not as expensive as the custom Muon sticker shock suggests, but it's still a pile of stuff to maintain in the forest.

Particle builds

Device	Role	Notes
Muon (Olivine-Wx1, Olivine-Wx2, bench units)	Weather + air quality	Temp, humidity, pressure, particulate sensing; can report over cellular and LoRaWAN.
Boron BRN404X (Bowman Wx1, bench Wx2/Wx3)	Weather	First field experiment on Bowman; cellular-only proved marginal → LoRaWAN parallel path.
Tachyon trailcam (Woods Cam 001)	Custom imaging (in progress)	Not the Vosker/GardePro units on the website today.

LoRaWAN builds

Device	Role	Sites / notes
RAK2560 + RK900-09	Full weather pack (temp, humidity, pressure, wind, battery)	Forest Lands Ridge
Dragino DDS75-LB	Snow depth	Ultrasonic distance to snow surface
Dragino S31B-LS	Temp / humidity / door	North Twin Approach, West Ridge, South Twin Approach, Baker View Turn, Dailey Prairie, N Twin Extra, etc.
Dragino D20S-LS	Soil temperature	Dailey Prairie ground sensor
RAK10701 Plus	Field tester	GPS / RF diagnostics when roaming

Other fused sources

Same backend also ingests **WeatherFlow Tempest**, **PurpleAir**, **OpenWeatherMap**, and a few other home and cron-fed sources — all into the same database (see §5).

5. Unified backend — Postgres / TimescaleDB

Everything — cellular, LoRaWAN, trail cams, fused external sources — lands in one pipeline:

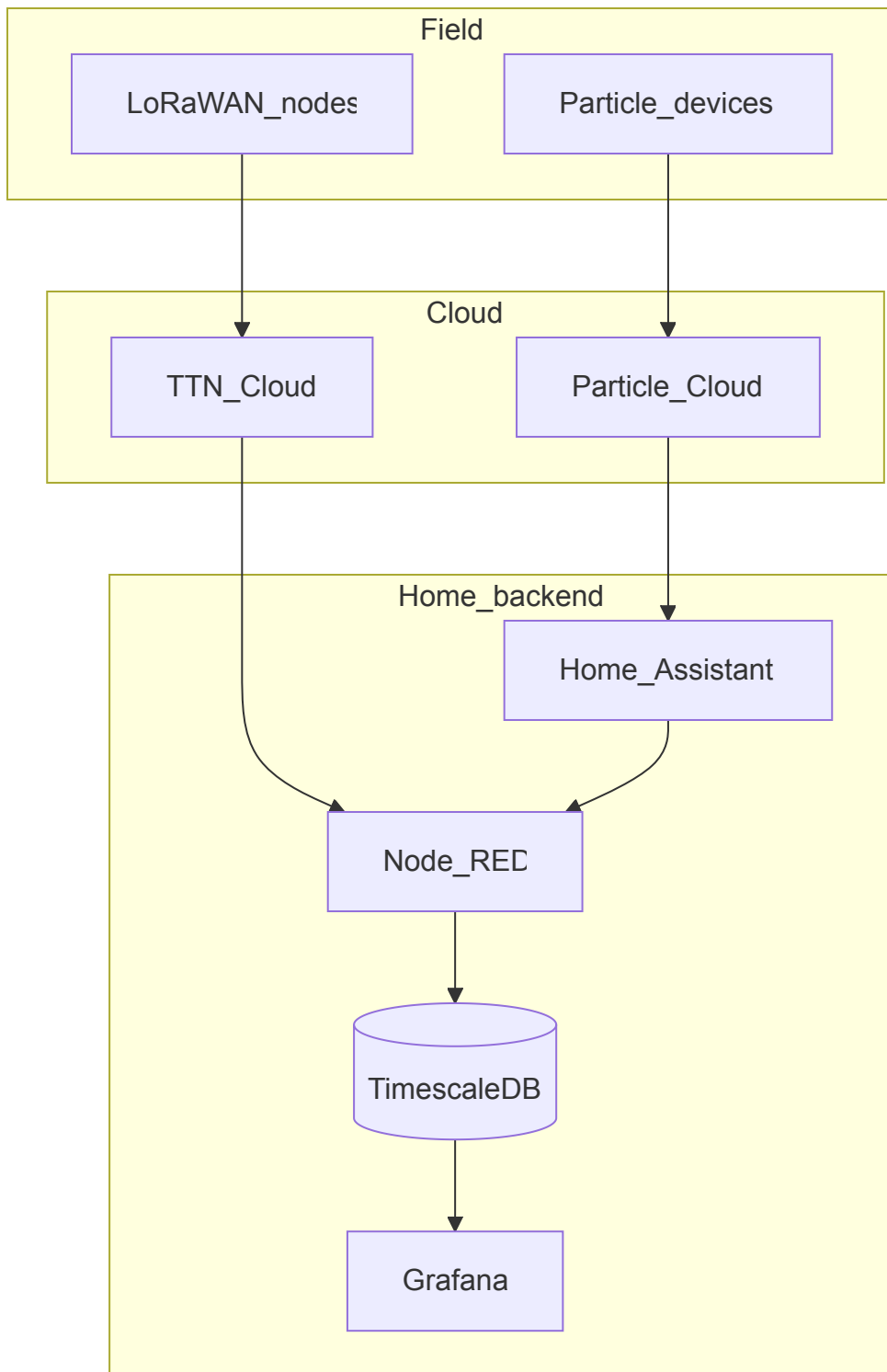
1. **TTN** or **Particle Cloud** → MQTT
2. Payload decode → **common flat JSON shape**
3. **Postgres / TimescaleDB** time-series storage
4. **Node-RED** for extraction, normalization, and publishing

TimescaleDB is the single time-series store.

Schema / table	Purpose
lorawan.*	Raw uplinks, gateway RX, aggregates, signal quality, device locations
weather.readings	Unified hypertable; source distinguishes LoRaWAN, Particle, WeatherFlow, PurpleAir, etc.

Grafana dashboards and **Windy** publishing consume the same database.

Diagram — field to database



6. Where the data lives (and what's missing)

Most weather data you can use today is on [Windy.com](https://www.windy.com) — Node-RED publishes the readings Windy accepts (temp, pressure, wind, etc.).

Not on Windy yet (in the database, working on public access):

- Air quality (AQI / particulate)
- Soil and ground / probe temperatures
- Some other specialty channels

Ideas for a good public home for those streams are welcome.

Trail camera images are on the project website — fed by the Vosker/GardePro scraper pipeline from §1. Thousands of frames so far, seasonal timelapse in the works.

7. Privacy, trail cameras, and being thoughtful in the woods

People go into the woods partly to **get away** from screens and surveillance. Random tech on a skin track is the last thing anyone needs. I get that.

- Cameras target **terrain and conditions**, not people.
- They have still captured a few passersby — **unintentionally**, and I'm sorry if that bothered anyone.
- **Automated filtering** drops images with people before anything goes public.
- **No faces have been posted.**
- Cameras are being **relocated** higher and farther from travel corridors where views stay useful.

If you find a QR-tagged box or a camera and have concerns, reach out via the site. I'd rather adjust placement or pull something than make the backcountry feel like a surveillance project. The intent is **safety and recreation** — conditions, smoke, wind, snow — not monitoring people.

8. How you can help (or just say hi)

Summer is maintenance season — heated batteries, gateway moves, wind sensor placements, camera relocations — and there's a lot of gear to track in the woods.

If you're interested:

- **Hauling gear** to North Twin, South Twin, or stashing at an agreed spot
- **Field trips** — always happy to meet people who want to adventure outdoors
- **Engineering, documentation, or data ideas** — especially for the non-Windy streams

People have asked how to **chip in financially**. A simple "buy me a coffee" link for ongoing costs — solar, batteries, cellular, Starlink, parts — is coming to the site.

If you just use the Windy stations or camera views to plan a day, that's enough. That's why I built it.
